THE COKERNEL OF A MAP FROM A MODULE OF FINITE TYPE INTO A FINITELY PRESENTED MODULE IS FINITELY PRESENTED

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The local description of a finitely presented module as a quotient of a finite direct sum of the structure sheaf by a submodule of finite type. Used at the start of the counted proof (source lines 1658-1661). Provides the free module and the finite type submodule in which the computation takes place.
- The finiteness of the extension in a short exact sequence whose outer terms are of finite type, and the finiteness of images of finite type modules. Author lemma lemma-extension-finite-type, whose statement is reproduced verbatim as a context span of this card (source lines 1071-1081). Shows that the submodule generated by the image of the given map is again of finite type.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a finitely presented $\mathcal{O}_X$-module. Let $\varphi : \mathcal{G} \to \mathcal{F}$ be a morphism of $\mathcal{O}_X$-modules. If $\mathcal{G}$ is finite type, then* $\mathrm{Coker}(\varphi)$ *is finitely presented.*

Lemma 0.2. *Let X be a ringed space. The image of a morphism of $\mathcal{O}_X$-modules of finite type is of finite type. Let $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ be a short exact sequence of $\mathcal{O}_X$-modules. If $\mathcal{F}_1$ and $\mathcal{F}_3$ are of finite type, so is $\mathcal{F}_2$.*

Proof. Locally on X, we can write $\mathcal{F} = \mathcal{O}_X^{\oplus n}/\mathcal{M}$, where $\mathcal{M} \subset \mathcal{O}_X^{\oplus n}$ is a finite type $\mathcal{O}_X$-submodule. Therefore $\mathrm{Im}(\varphi) = \mathcal{N}/\mathcal{M}$, where $\mathcal{N} \subset \mathcal{O}_X^{\oplus n}$ is a $\mathcal{O}_X$-submodule containing $\mathcal{M}$. The $\mathcal{O}_X$-module $\mathrm{Im}(\varphi)$ is of finite type because $\mathcal{G} \to \mathrm{Im}(\varphi)$ is surjective and $\mathcal{G}$ is finite type. By Lemma 0.2, $\mathcal{N}$ is finite type. Thus $\mathrm{Coker}(\varphi) = \mathcal{O}_X^{\oplus n}/\mathcal{N}$ is finitely presented. □

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